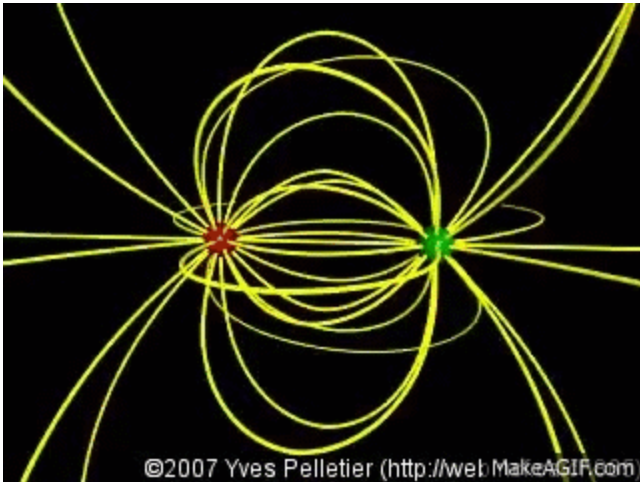


## 3\_popquiz electric field ↕

⚠ This is a preview of the published version of the quiz

Started: Mar 15 at 3:08pm

### Quiz Instructions



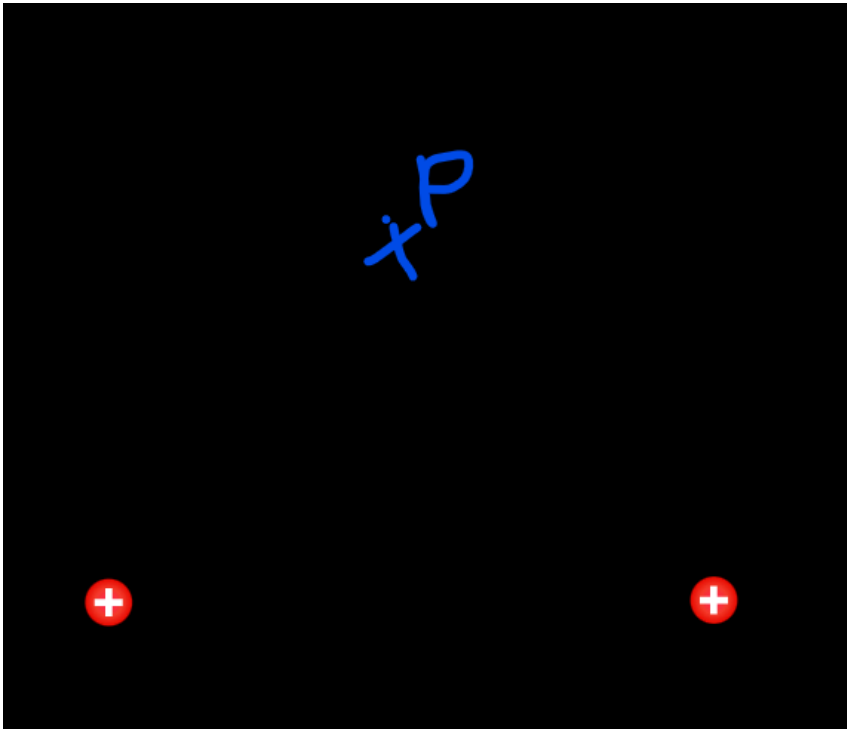
NOTE: an electron has a charge of  $-e$  and a proton has a charge of  $e$



Question 1 15 pts

Open this page: <https://phet.colorado.edu/en/simulations/charges-and-fields>   
(<https://phet.colorado.edu/en/simulations/charges-and-fields>)

Have the following charge distribution. Click on grid and click electric field. The point P is on the bisector of the line connecting the 2 charges. You can use the small sensor to visualize better, The net electric field  $E$  is pointing

☐

right

☐

left

☐

up

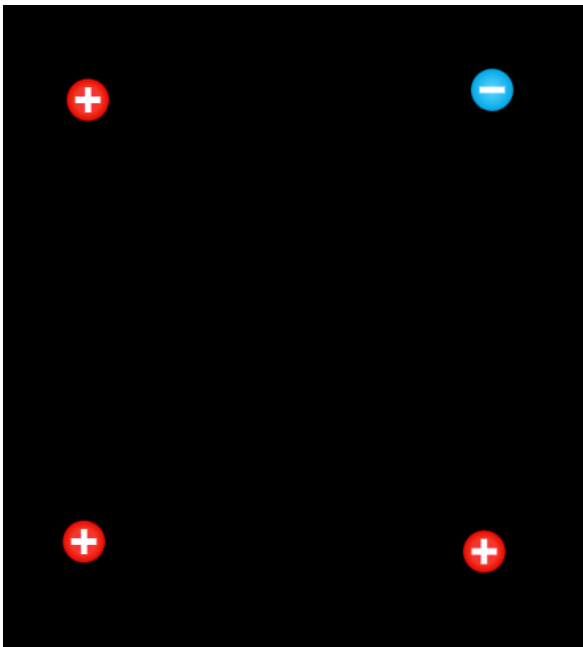
☐

down

☐

Question 2 20 pts

Same app.



Place the point P at the center of the parallelogram (formed by the charges). Use a sensor to find the direction of the net electric field. (you can also use adding vectors tail-to head), The net electric field is

☐

down

☐

up (along vertical)

☐

up and left (along diagonal)

☐

up and right (along diagonal)

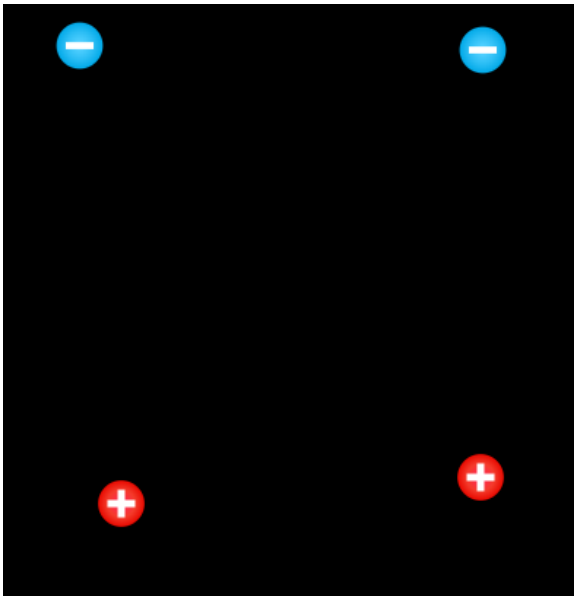
☐

down and left (along diagonal)

☐

Question 3 20 pts

Same app



The net electric field at the center of the parallelogram (formed by the charged)

☐

down and left (along diagonal)

☐

right

☐

up and right (diagonal)

☐

up along vertical

☐

down along vertical

☐

## Question 4 20 pts

COnsider a proton and an electron pinned at a distance  $d$  from each other. The ratio between the electric force between them and the gravitational force is about (order of magnitude)

☐ $10^{40}$ ☐ $10^{20}$ ☐

100

☐ $9 \cdot 10^9$ ☐

## Question 5 1 pts

open

[https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law\\_all.html](https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_all.html) 

([https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law\\_all.html](https://phet.colorado.edu/sims/html/coulombs-law/latest/coulombs-law_all.html))

and go to atomic scale (dealing forces between molecules). you have 2 molecules interacting with each other.

you see  $q_1 = -2e$  and  $q_2 = 4e$   $F_{12}$  is the force 1 on 2 and  $F_{21}$  is the force 2 on 1

What is true? (in magnitude)

☐ $F_{12} > F_{21}$ ☐ $F_{12} < F_{21}$ ☐ $F_{12} = F_{21}$ ☐ $F_{12} = F_{21} = 0$ ☐

## Question 6 20 pts

building on the previous question

$e$  is the elementary charge. So molecule 2 is missing

☐

4 protons

☐

it's neutral

☐

2 electrons

☐

4 electrons



Question 7 5 pts

building on previous question.

q2 (molecule2) has a charge of (in C)

hint: e is elementary charge

☐

4 C

☐ $-6.4 \cdot 10^{-19} \text{ C}$ ☐ $1.6 \cdot 10^{-19} \text{ C}$ ☐ $6.4 \cdot 10^{-19} \text{ C}$ 

Not saved

Submit Quiz